

SUBPHONETIC MODELING WITH MARKOV STATES — SENONE

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ABSTRACT

We will never have sufficient training data to model all the various acoustic-phonetic phenomena. How to capture important clues and estimate those needed parameters reliably is one of the central issues in speech recognition. Successful examples include subword models, fenones and many other smoothing techniques. In comparison with subword models, subphonetic modeling may provide a finer level of details. We propose to model subphonetic events with Markov states and treat the state in phonetic hidden Markov models as our basic subphonetic unit — *senone*. Senones generalize fenones in several ways. A word model is a concatenation of senones and senones can be shared across different word models. Senone models not only allow parameter sharing, but also enable pronunciation optimization. In this paper, we will report preliminary senone modeling results, which have significantly reduced the word error rate for speaker-independent continuous speech recognition.

1 INTRODUCTION

For large-vocabulary speech recognition, we will never have sufficient training data to model all the various acoustic-phonetic phenomena. How to capture important clues and estimate those needed parameters reliably is one of the central issues in speech recognition. To share parameters among different word models, context-dependent subword models have been used successfully in many state-of-the-art speech recognition systems [1, 2, 3, 4, 5]. The principle of parameter sharing can also be extended to subphonetic models. For subphonetic modeling, fenones [6, 7] have been used as the front end output of the IBM acoustic processor. To generate a fenonic pronunciation, multiple examples of each word were obtained. A fenonic word model is built from every utterance. The fenonic baseform (or pronunciation) for a word was chosen from the multiple fenonic word models so that it had the maximum joint likelihood for generating each of the examples. The codeword-dependent fenone models were then trained just like phonetic models. We believe that codeword-dependent fenones may be insufficient for large-vocabulary continuous speech recognition. In addition, the determination of a fenonic word model can not consider multiple examples simultaneously.

In this paper, we propose to model subphonetic events with Markov states. We will treat the state in hidden Markov models (HMMs) as a basic subphonetic unit — *senone*, which is named after fenone. The total number of HMM states in a system is often too large to be well trained. To reduce the amount of free parameters, we can cluster those state-dependent output distributions.

Here, a clustered output distribution is denoted as a senone. Conceptually, either a phone or word model is allowed to be formed by an arbitrary network of senones, and senones can be shared across different models as illustrated in Figure 1.

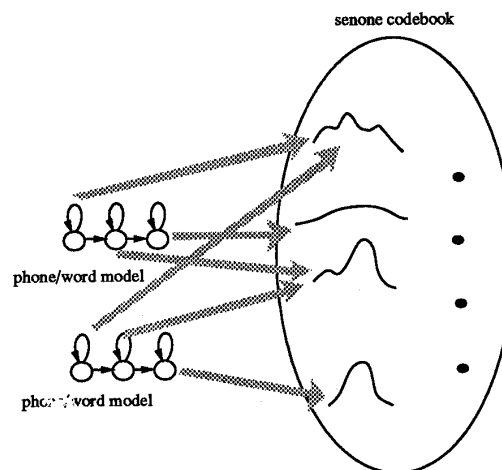


Figure 1: The senone hidden Markov model.

The advantages of senones include better parameter sharing and improved pronunciation optimization. After clustering, different states in different models may share the same senone if they exhibit acoustic similarity. Clustering at the granularity of the state rather than the entire model (like generalized triphones) can keep the dissimilar states of two models apart while the other corresponding states are merged, and thus lead to better parameter sharing. For instance, the first, or the second states of the /ey/ phones in PLACE and RELATION may be tied together. However, to magnify the acoustic effects of the right contexts, their last states may be kept separately. In addition to finer parameter sharing, senones also give us the freedom to use a larger number of states for each model. Although an increase in the number of states will increase the total number of free parameters, with senone sharing we can essentially eliminate those redundant states and have the luxury of maintaining the necessary ones.

Since senones depend on Markov states, the senonic baseform of a word can be constructed naturally with the forward-backward algorithm. Regarding pronunciation optimization, we can use the forward-backward algorithm to iteratively optimize a senone sequence appropriate for modeling multiple utterances of a word.

That is, given the multiple examples, we can train a word HMM with the forward-backward algorithm. When the reestimation reaches its optimality, the estimated states can be *quantized* with the codebook of senones. The closest one can be used to label the sequence of states of the word HMM. This sequence of senones becomes the senonic baseform of the word. Here arbitrary sequences of senones are allowed to provide the freedom for the automatically learned pronunciation. After the senonic baseform of every word is determined, the senones can be reestimated by the forward-backward algorithm. Although each word model generally has more states than the traditional phoneme-concatenated word model, the number of parameters remains the same since the size of the senone codebook is intact.

By constructing senone codebook and using senones in the triphone system, we were able to reduce the word error rate of the speaker-independent Resource Management task by 20% in comparison with the recently improved SPHINX system [8]. When senones are used for pronunciation optimization, our preliminary results gave us another 15% error reduction in a speaker-independent continuous spelling task. The word error rate was reduced from 11.3% to 9.6%.

2 PARAMETER SHARING

In phone-based HMM systems, each phonetic model is formed by a sequence of states. Phonetic models are shared across different word models. In fact, the state can also be shared across different phonetic models. This section will describe the usage of senones for parameter sharing.

2.1 Senone Construction

The number of triphones in a large vocabulary system is generally very large. With limited training data, there is no hope to obtain well-trained models. Therefore, different technologies have been studied to reduce the amount of parameters [1, 9, 2, 10, 11]. SPHINX used triphone clustering to reduce the number of triphone models [2]. In generalized triphones, every state of a triphone is merged with the corresponding state of another triphone in the same cluster. It may be true that some states are merged not because they are similar, but because the other states of the involved models resemble each other. To fulfill more accurate modeling, states with differently-shaped output distributions should be kept apart, even though the other states of the models are tied. Therefore, clustering should be carried out at the output-distribution level rather than the model level. The distribution clustering thus creates a senone codebook as Figure 2 shows [12]. The clustered distributions or senones are fed back to instantiate phonetic models. Thus, states of different phones may share the same senone. This is the same as the *shared-distribution model* (SDM) [13]. Moreover, different states within the same phone may also be tied together if too many states are used to model this phone's acoustic variations or if a certain acoustic event appears repetitively within the phone.

Senones also give us the freedom to use a larger number of states for each phonetic model. Although an increase in the number of states will increase the total number of free parameters, yet by clustering similar distributions we can essentially eliminate those redundant states and have the luxury to maintain the necessary ones [13].

1. All HMMs are first estimated.
2. Initially, every output distribution of all HMMs is created as a cluster.
3. Find the most similar pair of clusters and merge them together.
4. For each pair of clusters, consider moving every element from one to the other:
 - (a) move the element if the resulting configuration is an improvement;
 - (b) repeat until no such moves are left.
5. Go to step 3 unless some convergence criterion is met.

Figure 2: The construction of senones (clustered output distributions).

2.2 Performance Evaluation

We incorporated the above distribution clustering technique in the recently improved SPHINX system [8] and experimented on the speaker-independent DARPA Resource Management (RM) task with a word-pair grammar of perplexity 60. The test set consisted of the February 89 and October 89 test sets, totaling 600 sentences. Table 1 shows the word error rates of several systems.

System	Word Error	% Error Reduction
Improved SPHINX	4.7%	—
3500-SDM	4.2%	11%
4500-SDM	3.8%	20%
5500-SDM	4.1%	13%

Table 1: Results of the improved SPHINX vs. the SDM system.

In the improved SPHINX system, there were 1100 generalized triphones, each with 3 distinct output distributions. In the new system, we used 5-state Bakis triphone models and clustered all the output distributions in the 7500 or so triphones down to 3500–5500 senones. The system with 4500 senones has the best performance for the given 3990 training sentences. The similarity between two distributions was measured by their entropies. After two distributions are merged, the entropy-increase, weighted by counts, is computed:

$$(C_a + C_b)H_{a+b} - C_a H_a - C_b H_b$$

where C_a is the summation of the entries of distribution a in terms of counts, and H_a is the entropy. The less the entropy-increase is, the closer the two distributions are. Weighting by counts enables those distributions with less occurring frequency be merged before frequent ones. This makes each senone (shared distribution) more trainable.

2.3 Senone Codebook

To understand the quality of the senone codebook, we examined several examples in comparison with generalized triphone models. As shown in Figure 3, phone /ey/ in -PLACE and -LATION were mapped to the same generalized triphone. Similarly, phone /d/ in START and ASTORIA were mapped to another generalized triphone. Both has the same left context, but different right contexts. States with the same color were tied to the same senone in the 4500-SDM system. x , y , z , and w represent different senones. Figure (a) demonstrates that distribution clustering can keep dissimilar states apart while merging similar states of two models. Figure (b) shows that redundant states inside a model can be squeezed. It also reveals that distribution clustering is able to learn the same effect of similar contexts (/aa/ and /ao/) on the current phone (/d/).

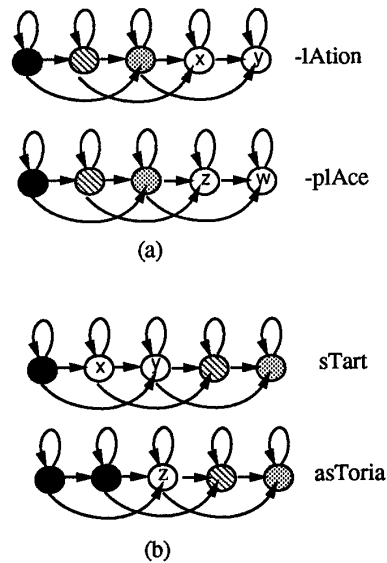


Figure 3: Examples of triphone-clustering and distribution-clustering. Figure (a) shows two /ey/ triphones which were in the same generalized triphone cluster; (b) shows two /d/ triphones in another cluster. States with the same color were tied to the same senone. x , y , z , and w represent different senones.

It is also interesting to note that when 3 states, 5 states, and 7 states are used with a senone codebook size of 4500, the average number of distinct senones in a triphone is 2.929, 4.655, and 5.574 respectively. This might imply that 5 states per phone are optimal to model the acoustic variations within a triphone unit for the given DARPA RM training database. In fact, 5-state models indeed gave us the best performance.

3 PRONUNCIATION OPTIMIZATION

As shown in Figure 1, senones can be shared not only by different phonetic models, but also by different word models. This

section will describe one of the most important applications of senones: word pronunciation optimization.

3.1 Senonic Baseform

Phonetic pronunciation optimization has been considered by [14, 15]. Subphonetic modeling also has a potential application to pronunciation learning. Most speech recognition systems use a fixed phonetic transcription for each word in the vocabulary. If a word is transcribed improperly, it will be difficult for the system to recognize it. There may be quite a few improper transcriptions in a large vocabulary system for the given task. Most importantly, some words may be pronounced in several different ways such as THE (/dh ax/ or /dh ih/), TOMATO (/t ax m ey dx ow/ or /t ax m aa dx ow/), and so on. We can use multiple phonetic transcriptions for every word, or to learn the pronunciation automatically from the data.

We want to find out the most appropriate senonic baseform for a given word when training examples are available. We first compute the average duration of each word, and build a Bakis word HMM with the number of states equal to a predefined portion of the duration. Using the forward-backward algorithm, we can estimate the word HMM with the given multiple examples. When the forward-backward optimization is completed, each estimated state is quantized with the senone codebook. We can thus obtain a sequence of senones that is most suitable to model the given set of training tokens. Here arbitrary sequences of senones are allowed to provide the freedom for the automatically learned pronunciation. This senonic baseform tightly combines the model and acoustic data. After the senonic baseform of every word is determined, the senones can be re-trained.

In comparison with fenones, senones take full advantage of the multiple utterances in baseform construction. The senone codebook also has a better acoustic resolution in comparison with the 200 VQ-dependent fenones. In addition, both phonetic baseform and senonic baseform can be used together. So we can keep using phonetic baseform when training examples are unavailable. Although each senone-word model generally has more states than the traditional phoneme-concatenated word model, the number of parameters are not increased since the size of the senone codebook is fixed.

3.2 Performance Evaluation

As a pivotal experiment for pronunciation learning, we used the speaker-independent continuous spelling task (26 English alphabet). No grammar is used. There are 1132 training sentences from 100 speakers and 162 testing sentences from 12 new speakers. The training data were segmented into words by a set of existing HMMs. For each word, we split its training data into several groups by a DTW clustering procedure according to their acoustic resemblance. Different groups represent different acoustic realizations of the same word. For each word group, we estimated the word model and computed a senonic baseform as the previous subsection describes. The number of states of a word model was equal to 75% of the average duration. The Euclidean distance was used as the distortion measure in senone quantization.

We calculated the predicting ability of the senone-word model

$M_{w,g}$ obtained from the g -th group of word w as:

$$\sum_{\mathbf{X}_w \notin \text{group } g} \log P(\mathbf{X}_w | M_{w,g}) / \sum_{\mathbf{X}_w \notin \text{group } g} |\mathbf{X}_w|$$

where $\mathbf{X}_w$ is an utterance of word w .

For each word, we picked two models that had the best predicting abilities. The pronunciation of each word utterance in the training set was labeled by:

$$\text{model}(\mathbf{X}_w) = \underset{M_{w,g} \in \text{top2}}{\operatorname{argmax}} \{P(\mathbf{X}_w | M_{w,g})\}$$

After the training data were labeled in this way, we re-trained the system parameters by using the senonic baseform. Table 2 shows the word error rate. Both systems used the sex-dependent semi-continuous HMMs. The baseline used word-dependent phonetic models. Therefore, it was essentially a word-based system. Fifty-six word-dependent phonetic models were used. Note both systems used exactly the same amount of parameters.

System	Word Error	% Error Reduction
phonetic baseform	11.3%	—
senonic baseform	9.6%	15%

Table 2: Results of the phonetic baseform vs. the senonic baseform.

This preliminary results indicate that senonic baseform can capture detailed pronunciation variations for speaker-independent speech recognition.

4 CONCLUSION

In this paper, we developed the framework of senones — state-dependent subphonetic unit. A phonetic model or a word model can be constructed by quantizing its states with the senone codebook. We can reduce the the amount of system parameters with the senone codebook, which renders finer acoustic modeling and provides a way to learn the model topology. In the mean time, we can construct senonic baseform to improve phonetic baseform without enlarging system parameters. We demonstrated that senones can significantly improve speaker-independent continuous speech recognition. However, the experiment on senonic baseform was very preliminary. We will conduct large-vocabulary senonic baseform experiments in the future.

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